

USTHB · Faculty of Computer Science · 2025/2026

Department of AI and Data Science

Realization of an Intelligent Supermarket with Connected Carts

and a Personalized Recommendation System

The logo for CartyGo is displayed within a dark blue rectangular box. The text "CartyGo" is written in a light blue, sans-serif font, with the "C" and "G" being slightly larger and bolder than the other letters.

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Presentation Plan

01	Introduction	Context, problems faced by Algerian supermarkets
02	Objectives	Goals of the CartyGo project
03	Key Definitions	IoT, Smart Supermarket, Connected Cart, Recommendation Systems
04	Proposed Solution	CartyGo overview — actors, interfaces and interactions
05	Conception	Use case diagrams, sequence diagrams, class diagram
06	Recommendation System	Hybrid approach: collaborative + content-based filtering
07	Technical Architecture	Tech stack, microservices, database, security
08	Conclusion & Perspectives	Achievements and future improvements

Introduction

In a world where technology evolves very fast, digitalization has transformed many sectors, and retail is no exception. Around the world, supermarkets increasingly rely on modern digital tools to improve both the shopping experience and the management of the store.

In **Algeria**, however, most supermarkets still operate mainly with traditional and manual methods. Despite the growth of the sector and the rising demand for more modern solutions, this gap remains.

This raises several key challenges...

Identified Problems

For the Customer

- Time lost searching for products in aisles
- Long queues at checkout
- No personalized recommendations
- No real-time budget tracking
- No intelligent cart management

For the Manager

- Limited visibility on stock & expiry dates
- No link between in-store experience and internal operations

Conclusion

No complete local solution exists to digitalize and improve the in-store shopping experience.

◆ GOALS

Project Objectives

■ Pleasant Customer Experience	Mobile app synchronized in real time with the connected cart
■ Seamless Barcode Management	From scan capture to instant transmission to the smartphone
■ Purchase Transparency	Instant display of cart content and running total
■ Personalized Recommendations	Using purchase history, health profile and allergies
■ Cross-selling & Promotions	Similar-item suggestions and targeted promotion alerts
■ Manager Dashboard	Real-time monitoring of stock and commercial performance

Key Definitions

Internet of Things (IoT)

A technological ecosystem integrating hardware and software. Connects physical objects to a network enabling data exchange with little or no human intervention. Powered by AI, cloud, big data, low-cost sensors and resilient connectivity.

Smart Supermarket

A self-service supermarket integrating advanced technologies: AI, sensors, cameras, electronic labels, mobile apps. Models include: self-checkout stores, cashier-less autonomous stores, smart shelves, and supermarkets with connected carts.

Connected Cart

A cart equipped with cameras, sensors, barcode/QR/Rfid readers and a screen — connected to the store system and the customer's mobile app. Three functions: data collection & analysis, personalization & recommendations, and integrated payment.

Recommendation System

A software solution that suggests the most relevant products to a user based on their profile and purchase history. Three types: Content-based, Collaborative, and Hybrid. CartyGo uses a Hybrid approach.

◆ OUR SOLUTION

CartyGo — System Overview

A centralized digital system linking the three actors of the supermarket: **Customer**, **Manager**, and **Cashier**. Real-time communication and constant synchronization between the mobile app, the connected cart, and the web dashboard.

Customer (Mobile App)	Manager (Web Dashboard)	Cashier (Web Dashboard)
• Browse products • Track cart & total live • Receive recommendations • Pay from smartphone • Scan via cart tablet	• Real-time store overview • Sales & performance stats • Manage products & promotions • Stock monitoring & alerts	• Confirm cash purchases • Verify transactions • Validate payment conformity

Use Case: Product Management (Client)

After registration, the customer provides personal info and selections (preferred categories, health profile, allergies). After authentication, they can browse the catalogue, search products, and scan items via the smart cart. Each scanned product is added to the virtual cart.

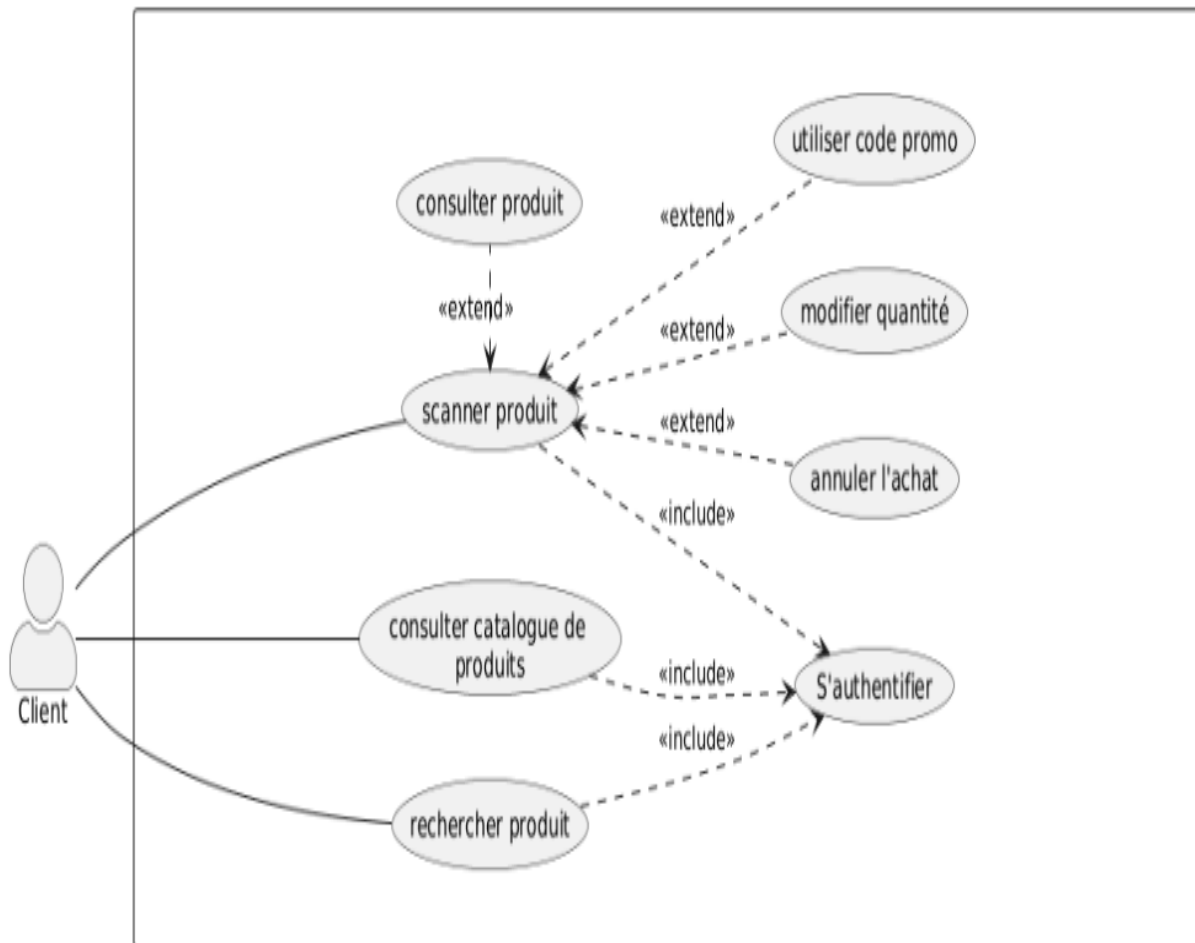


Fig. 6.1 — Use Case: Product Management

Use Case: Payment & Purchase Confirmation

To finalize, the customer chooses a payment method: electronic (bank card via Chargily Pay) or cash. The system applies any promotions or discount codes set by the manager.

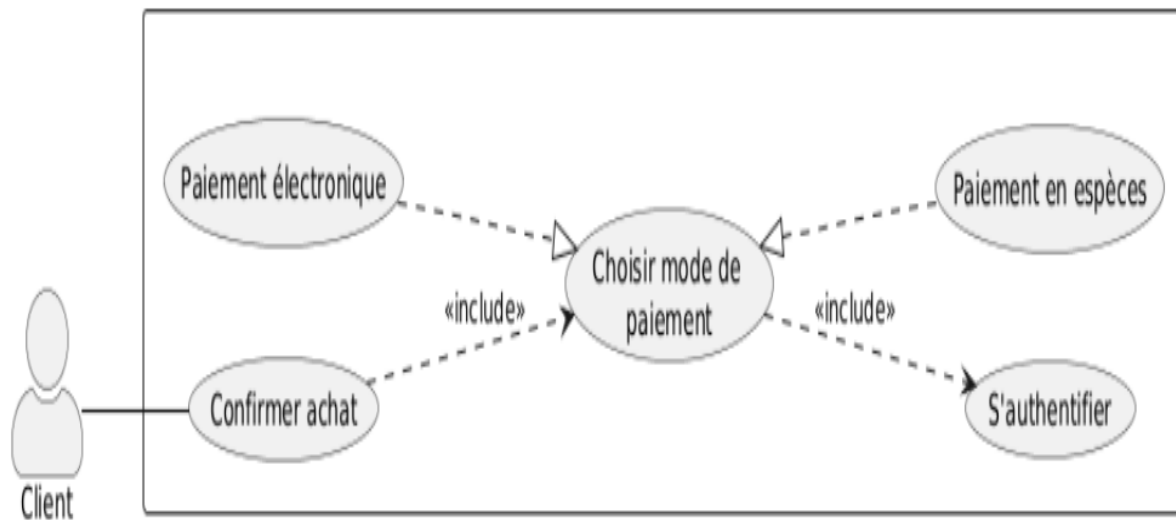


Fig. 6.2 — Use Case: Payment and Purchase Confirmation

Use Case: Products & Promotions Management (Manager)

After authentication, the manager manages products (add, modify, delete) and creates/manages promotions. Goal: stimulate sales and keep the catalogue up to date.

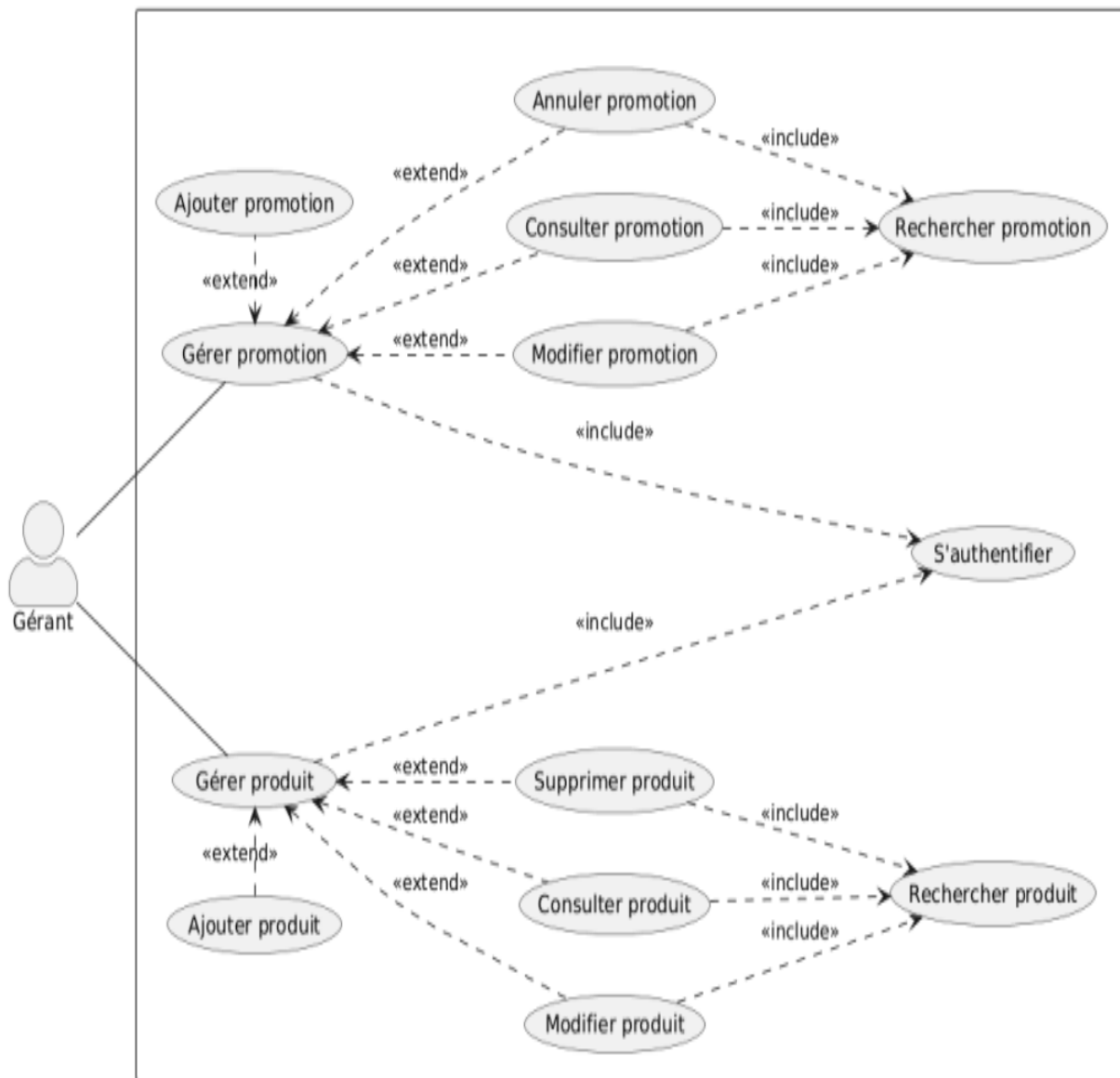


Fig. 6.4 — Use Case: Products and Promotions Management

Use Case: Cash Payment Management (Cashier)

The cashier intervenes mainly in the cash payment process. They validate transactions made at the till and verify payment conformity before final confirmation in the system.

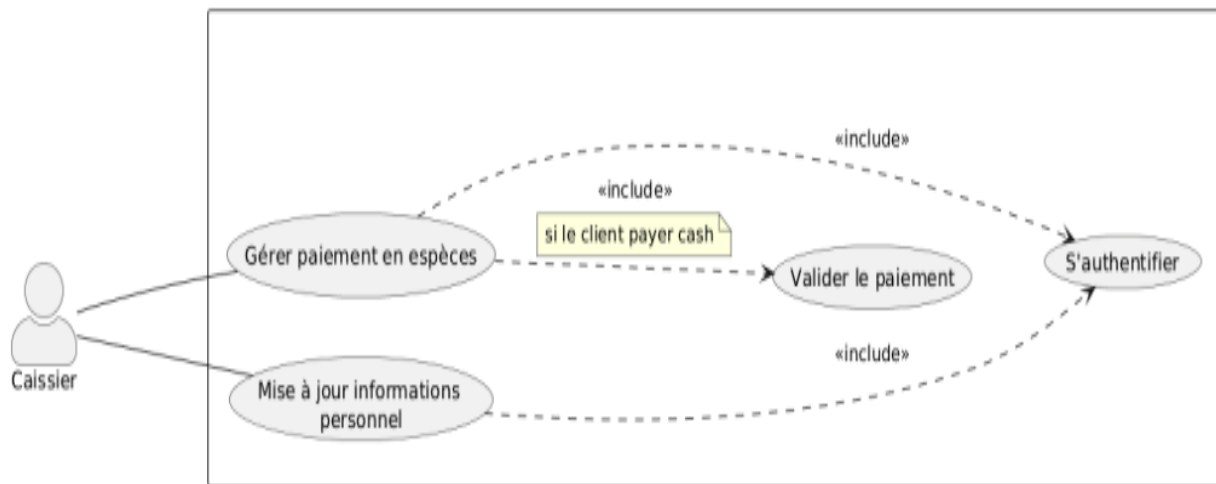


Fig. 6.6 — Use Case: Cash Payment Management

Sequence Diagram: Purchase Process

After authentication, the customer scans product barcodes in a loop. Each scan triggers a database request to fetch product info. The cart content is synchronized in real time with the user profile. Once finished, the final list is shown and the payment module is activated.

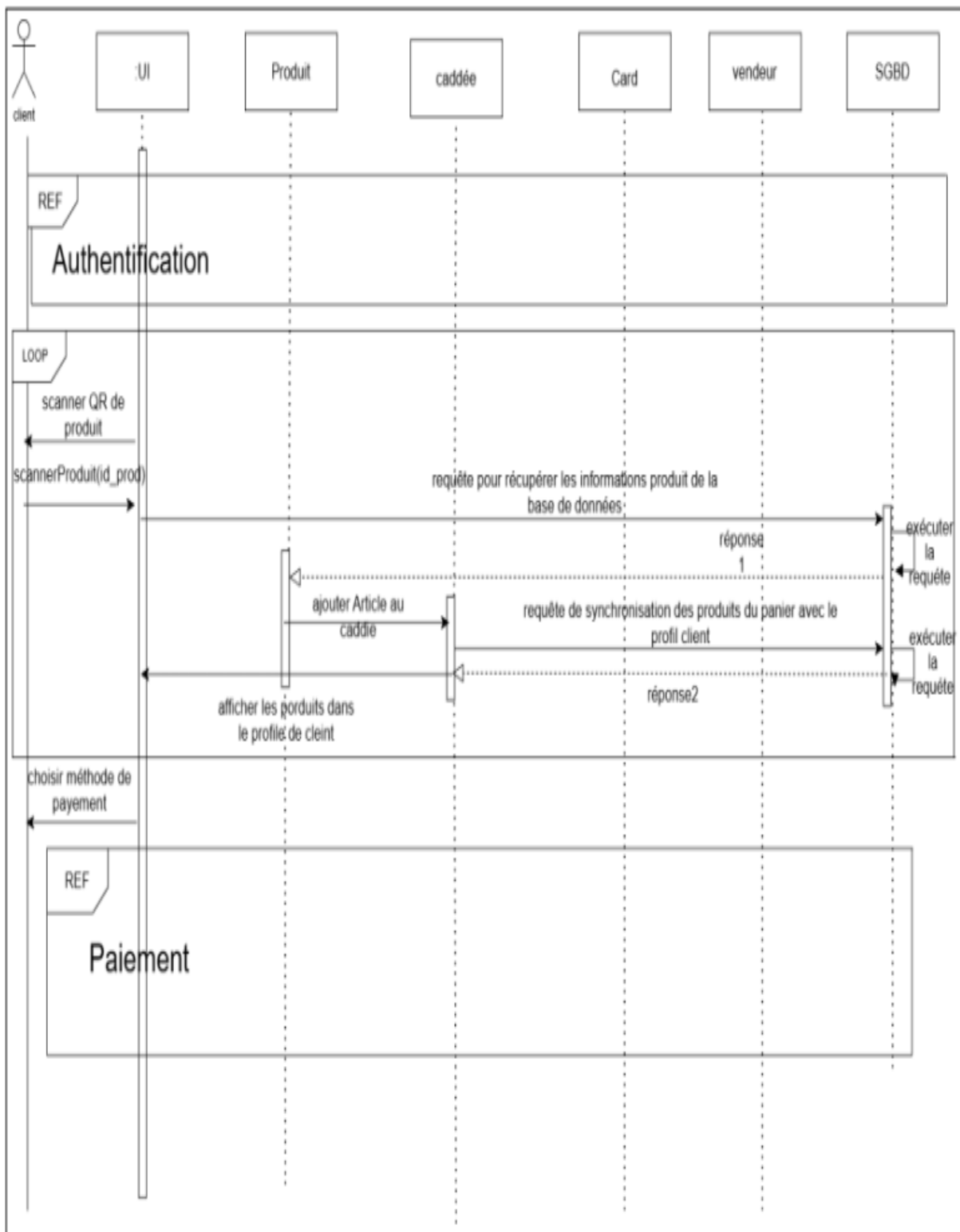
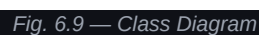


Fig. 6.7 — Sequence Diagram: Purchase Process

Class Diagram

Core classes: Customer, Order, OrderItem, Product, Cart, Wagon, Staff, Cashier, Manager, Promotion, Feedback — with enumerations for roles and statuses.



Hybrid Recommendation System

Approach: Collaborative Filtering + Content-Based Filtering

Step	Description
1. Data Collection	Purchase history, ratings, preferred categories, allergies, health profile
2. Collaborative Filtering	Build USER-PRODUCT matrix → compute Jaccard similarity → KNN (K=10 nearest users) → score candidate products
3. Content-Based Filtering	Filter products incompatible with health profile (■ red) or containing allergens (■ yellow) → compute ScoreCB from preferred categories
4. Final Score	$\text{Score}_{\text{final}}(p) = \alpha \times \text{Score}_{\text{CF}}(p) + (1-\alpha) \times \text{Score}_{\text{CB}}(p)$ with $\alpha = 0.7$
5. Cold-Start Fix	New users without history are served recommendations based solely on their profile (health, allergies, categories)

Key Formulas

Jaccard Similarity: $\text{Sim}(A,B) = |A \cap B| / |A \cup B|$

CF Score: $\text{Score}_{\text{CF}}(p) = \sum \text{Sim}(i,u) \cdot v(u,p) / \sum \text{Sim}(i,u)$ [threshold ≥ 0.6]

Final Score: $\text{Score}_{\text{final}} = 0.7 \times \text{Score}_{\text{CF}} + 0.3 \times \text{Score}_{\text{CB}}$

System Architecture

Component	Technology	Role
Main API	Node.js + Express	Auth, profiles, products, orders, payments (MVC)
IoT Server	Socket.IO	Real-time cart ↔ phone synchronization
Recommendation	Python + FastAPI	Hybrid collaborative + content-based filtering
Web Dashboard	Next.js + React	Manager & cashier interface (SSR)
Mobile App	Flutter / Dart	Customer-facing Android/iOS application
Cart Tablet App	Flutter / Dart	Barcode scanner + real-time sync
Database	MongoDB Atlas	Shared NoSQL cloud database (all services)
Payment	Chargily Pay	CIB + Dahabia payment integration
Security	JWT + bcrypt + OTP	Authentication, hashing, password reset

Conclusion & Perspectives

What We Achieved

- A complete smart-supermarket ecosystem with three interconnected interfaces
- A hybrid, personalized recommendation system
- Real-time synchronization between the connected cart and the mobile app
- A centralized dashboard for stock and commercial performance

Future Perspectives

- Computer vision for automatic product detection (no manual scan)
 - Deep learning model to refine recommendation accuracy
 - Deployment on real hardware (Raspberry Pi, RFID readers, embedded tablets)
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Thank you for your attention

Questions & Discussion